

Ironwood, the core

how Zcash verifies a shielded transfer without seeing it

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What a validator checks

1. the coin **exists** — look it up
2. the spend is **authorised** — script and signature
3. it is not spent **twice** — delete it
4. value is **conserved** — $\sum \text{in} \geq \sum \text{out}$

Every check reads the transaction in the clear: which coin, whose key, how much.

Privacy, precisely

Hide the sender, the receiver and the amount — while a stranger still runs all four checks.

check	mechanism
exists	a commitment in a Merkle tree, membership proved in zero knowledge
not twice	a nullifier: a keyed serial number, revealed once
conserved	homomorphic value commitments and a binding signature
authorised	a proof of key knowledge and a re-randomised signature

One zero-knowledge statement per Action carries all four; one proof per bundle.

Anchors

Arithmetic with a modulus

For a prime p , the field $\mathbb{F}_p$ is arithmetic modulo p .

$$\mathbb{F}_{97} = \{0, 1, \dots, 96\}, \quad 96 + 2 = 1, \quad 5 \cdot 39 = 1.$$

Every nonzero element has an inverse: $5^{-1} = 39$ in $\mathbb{F}_{97}$.

Addition, subtraction, multiplication, division by nonzero elements. The protocol's fields have $p \approx 2^{254}$; the toy $\mathbb{F}_{97}$ returns in the proof.

A group and a hard problem

$$\mathcal{E}(\mathbb{F}_p) : \quad y^2 = x^3 + 5 \quad \text{over } \mathbb{F}_p, \quad p \approx 2^{254}, \quad |\mathcal{E}| = q \text{ prime}$$
$$[a] G := \underbrace{G + G + \cdots + G}_{a \text{ times}} \quad \text{given } G \text{ and } [a] G, \text{ find } a : \text{ infeasible}$$

$G, [a] G \in \mathcal{E}$ points; $a \in \mathbb{F}_q$ a scalar; both fields $\approx 2^{254}$.

- Pallas: points over $\mathbb{F}_p$, scalars in $\mathbb{F}_q$.
- Vesta: the same curve with p and q swapped — a cycle.
- Everything below is built from $[a] G$ and the discrete log.

Commitments

$$C(v, r) := [v] V + [r] R \quad V, R \text{ independent points}$$

$$v, r \in \mathbb{F}_q; V, R, C \in \mathcal{E}.$$

- **Hiding:** $r \mapsto [r] R$ permutes the group, so $[v] V + [r] R$ has the same distribution for every v .
- **Binding:** two openings of one C give $[v_0 - v_1] V = [r_1 - r_0] R$, hence the forbidden $\lambda = (v_0 - v_1)(r_1 - r_0)^{-1}$ with $V = [\lambda] R$.
- **Homomorphic:** $C(v_1, r_1) + C(v_2, r_2) = C(v_1 + v_2, r_1 + r_2)$.

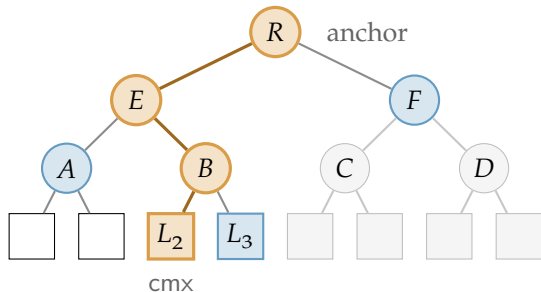
A sealed envelope that can be added to other sealed envelopes.

Hashes and Merkle trees

$$B = H(L_2, L_3), \quad E = H(A, B), \\ R = H(E, F) = \text{anchor}$$

leaves and nodes in $\mathbb{F}_p$.

- Nobody can find two inputs with one output.
- Position 010_2 ; siblings (L_3, A, F) . Depth 32 in the protocol.
- The tree is public; the leaf and the path stay hidden.
- Append only: old roots stay valid.



A sigma protocol: Schnorr identification

Prover knows x with $X = [x] G$.

Prover

Verifier

$R = [k] G, k \text{ random}$

$c \text{ random}$

$s = k + c x$

Verifier checks $[s] G \stackrel{?}{=} R + [c] X$

$x, k, c, s \in \mathbb{F}_q; G, X, R \in \mathcal{E}$.

Why it proves, and why it hides

Special soundness. Two accepting transcripts with the same R :

$$(R, c, s), (R, c', s') \implies x = \frac{s - s'}{c - c'}$$

whoever answers two challenges knows x .

Zero knowledge. A simulator with no x picks c and s first:

$$R := [s] G - [c] X$$

the transcript has exactly the real distribution. The verifier learns nothing it could not have made itself.

Fiat–Shamir: the verifier becomes a hash

$$c := H(X, R, m)$$

- The prover cannot pick R after c : c depends on R .
- The pair (R, s) is now a signature on m — Schnorr.
- **sigma protocol + Fiat–Shamir = Schnorr signature.**
- Zcash's RedPallas is Schnorr on Pallas, with one addition: the key can be re-randomised (later).

Zero-knowledge proofs, in general

A relation $\mathcal{R}(x, w)$: public statement x , private witness w .

$$\text{Prove}(x, w) \rightarrow \pi$$

$$\text{Verify}(x, \pi) \in \{0, 1\}$$

- **Complete:** an honest π verifies.
- **Knowledge sound:** from any convincing prover an extractor pulls a w with $\mathcal{R}(x, w)$.
- **Zero knowledge:** a simulator given only x reproduces π .
- **SNARK:** π short (logarithmic) and non-interactive (Fiat–Shamir).

Schnorr is the proof for $\mathcal{R} = \{(X, x) : X = [x] G\}$.

A polynomial cannot lie everywhere

$$D \neq 0, \deg D \leq d \implies \Pr_{z \leftarrow \mathbb{F}}[D(z) = 0] \leq \frac{d}{|\mathbb{F}|}$$

$D \in \mathbb{F}[X]$ a polynomial over a field $\mathbb{F}$; $z \in \mathbb{F}$.

A nonzero polynomial has at most d roots. Fix D first, draw z second: one random probe catches it.

With $|\mathbb{F}| \approx 2^{254}$ and $d \approx 2^{14}$ the odds of escape are about 2^{-240} .

The core of Orchard

A note

$$n = (d, \text{pk}_d, v, \rho, \psi, \text{rcm})$$

$$m = g_d^* \parallel \text{pk}_d^* \parallel \text{LE}_{64}(v) \parallel \text{LE}_{255}(\rho) \parallel \text{LE}_{255}(\psi),$$
$$\text{cm} = \text{NoteCommit}_{\text{rcm}}(m), \quad \text{cmx} = \text{Extract}(\text{cm}) \in \mathbb{F}_p$$

$$d \in \{0, 1\}^{88}; \text{pk}_d, g_d, \text{cm} \in \mathcal{E}; v \in [0, 2^{64}); \rho, \psi \in \mathbb{F}_p; \text{rcm} \in \mathbb{F}_q; \text{cmx} \in \mathbb{F}_p.$$

(d, pk_d) the recipient's diversified address

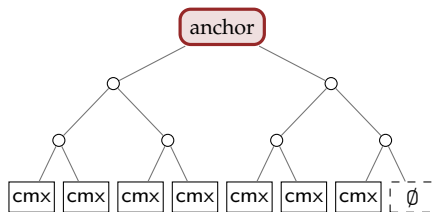
v the value, $0 \leq v < 2^{64}$

ρ, ψ uniqueness seed and nullifier salt

rcm the commitment trapdoor

On chain: cmx and a ciphertext. Nothing else about the note.

The chain's shielded state



- **The commitment tree:** depth 32, append only; every block's final root is an *anchor*.
- **The nullifier set:** the spent half, growing; a nullifier names no leaf.
- A spend proves membership under some anchor and reveals a nullifier.

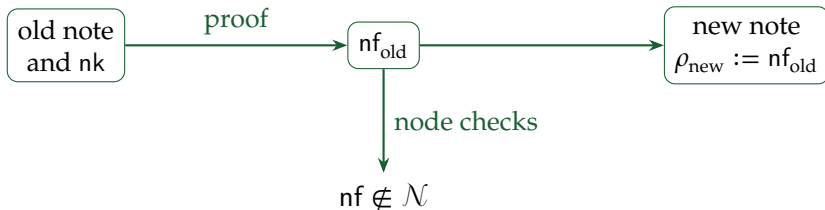
The nullifier

$$\text{nf} := \text{Extract}([F_{\text{nk}}(\rho) + \psi] G^{\text{nf}} + \text{cm})$$

$$\text{nk}, \rho, \psi, F_{\text{nk}}(\rho) \in \mathbb{F}_p; G^{\text{nf}}, \text{cm} \in \mathcal{E}; \text{nf} \in \mathbb{F}_p.$$

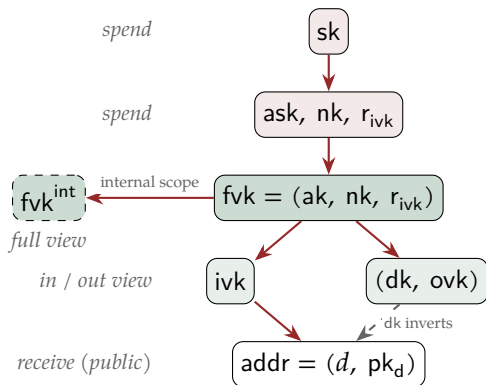
- Keyed by nk , the owner's nullifier key; F_{nk} is Poseidon, a hash the circuit evaluates natively.
- Seeded by ρ , salted by ψ , bound to cm .
- Chaining $\rho_{\text{new}} := \text{nf}_{\text{old}}$: every note inherits the nullifier of the note spent beside it, so ρ is unique.
- The sum is taken in $\mathbb{F}_p$, then read as a scalar ($p < q$).
- Unique, and unlinkable to cm without nk .

The proof and the node check different things



The set $\mathcal{N}$ holds every spent nullifier. Reject repeats; insert only when the transaction is accepted.

Keys



$ask, nk, r_{ivk} := \text{PRFexpand}_{sk}(06), (07), (08)$

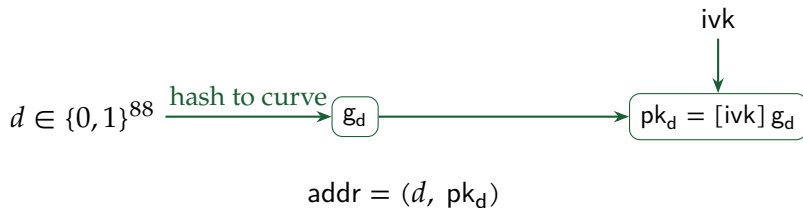
$ak := [ask] G^{\text{SpendAuth}}$

$ivk := \text{Commit}_{r_{ivk}}^{ivk}(\text{Extract}(ak), nk)$
 $(dk, ovk) := \text{PRFexpand}_{r_{ivk}}(82 \parallel \text{Extract}(ak) \parallel nk)$

$d := \text{FF1}_{dk}(j), g_d := \text{GroupHash}(d),$
 $pk_d := [ivk] g_d$

$ask, r_{ivk} \in \mathbb{F}_q; nk \in \mathbb{F}_p; ak \in \mathcal{E}; ivk \in \mathbb{F}_p$, used as a scalar ($p < q$); dk, ovk byte strings;
 $d \in \{0, 1\}^{88}; pk_d \in \mathcal{E}.$

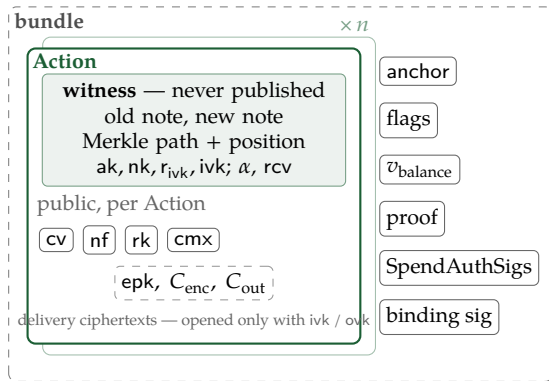
Diversified addresses



$d \in \{0, 1\}^{88}; g_d, pk_d \in \mathcal{E}; ivk \in \mathbb{F}_p$ as a scalar.

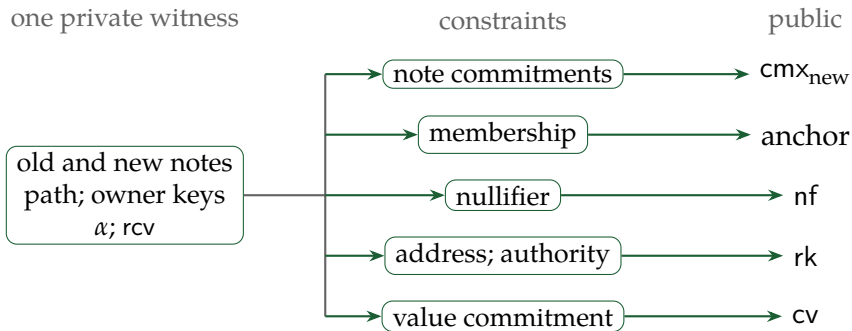
- Some 2^{88} addresses per key, all paid into one wallet.
- Unlinkable to each other, even by the payer: two addresses share ivk only as a discrete log.
- One ivk detects every payment to every address.

The Action



- One Action spends one note and creates one note.
- A dummy note pads either side; every Action looks alike.
- One proof per bundle covers all its Actions.

The statement, not the secret



Every constraint uses the same notes and the same owner keys.

The statement

check	clause	the proof establishes
exists	A3	a Merkle path from cm_{old} to the anchor
not twice	A5	the nullifier nf is this note's serial, keyed by nk
conserved	A4	the commitment cv hides $v_{old} - v_{new}$, both $< 2^{64}$
authorised	A6, A7	the key rk disguises ak ; ak, nk sit behind the address
well-formed	A1, A2	both commitments open to notes; $\rho_{new} = nf_{old}$
gated	A8–A10	spends, outputs, cross-address payments switched by flags

The witness: both notes, the path and position, the keys, α , rcv .

The one check the proof cannot make: has this nf been seen before.

Conservation is a cancellation

$$cv_i = [v_{\text{old},i} - v_{\text{new},i}] V + [rcv_i] R \quad \text{per Action}$$

Let b be the public net value leaving the pool (v_{balance}). The public binding key:

$$bvk := \sum_i cv_i - [b] V = \left[\underbrace{\sum_i (v_{\text{old},i} - v_{\text{new},i})}_{\Delta: \text{ imbalance}} - b \right] V + \left[\sum_i rcv_i \right] R$$

$v \in [0, 2^{64})$; $rcv_i \in \mathbb{F}_q$; $cv_i, bvk, V, R \in \mathcal{E}$; b a signed 64-bit integer; $\Delta \in \mathbb{F}_q$.

For a fully shielded transfer, b is the fee.

Why the binding signature enforces balance

A signature under bvk on base R proves knowledge of k with $\text{bvk} = [k] R$. With $s := \sum_i \text{rcv}_i$:

$$[\Delta] V = [k - s] R$$

$$k, s, \Delta, \lambda \in \mathbb{F}_q.$$

- Balanced: $\Delta = 0$, and the author knows $k = s$.
- Unbalanced: $\Delta \neq 0$ gives $\lambda = \Delta (k - s)^{-1}$ with $V = [\lambda] R$, the forbidden relation.

Schnorr again: a proof of knowledge of a discrete log, of a key that exists only when the books balance.

Field equality is not yet integer equality

$$q - 1 = -1, \quad q = 0 \quad \text{in } \mathbb{F}_q$$

So the proof also constrains every amount:

$$0 \leq v < 2^{64}$$

Bounded amounts and a bounded number of Actions keep an integer imbalance from wrapping around modulo q .

Authorisation without a fixed key

$$ak = [ask] G^{\text{SpendAuth}}, \quad pk_d = [ivk] g_d,$$

$$ivk = \text{Commit}_{r_{ivk}}^{ivk}(\text{Extract}(ak), nk)$$

$$rk := ak + [\alpha] G^{\text{SpendAuth}} = [rsk] G^{\text{SpendAuth}}, \quad rsk := ask + \alpha$$

$ask, \alpha, rsk \in \mathbb{F}_q$; $ak, rk \in \mathcal{E}$; $ivk, nk \in \mathbb{F}_p$.

- One ak per wallet; publishing it would stamp every spend.
- Fresh α per spend: rk is a uniform point, unlinkable.
- The signature is Schnorr under rk ; the circuit proves rk was formed from the ak behind the spent note's address.

Delivery

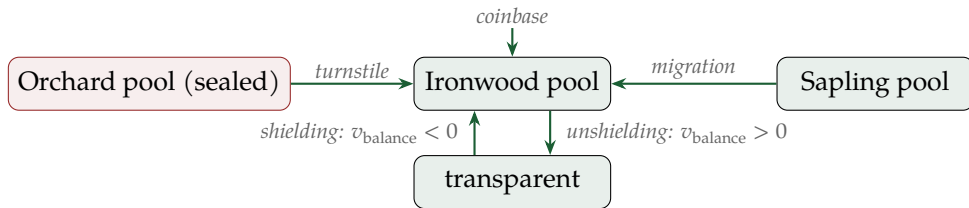
$$\text{epk} := [\text{esk}] g_d, \quad s := [\text{esk}] \text{pk}_d = [\text{ivk}] \text{epk}, \quad K := \text{KDF}(s, \text{epk})$$

$$C_{\text{enc}} := \text{AEAD}_K(0x03, d, v, \text{rseed}, \text{memo})$$

$\text{esk} \in \mathbb{F}_q; g_d, \text{pk}_d, \text{epk}, s \in \mathcal{E}; K \in \{0, 1\}^{256}; C_{\text{enc}}: 580 \text{ bytes}.$

- One-shot Diffie–Hellman to the diversified base, then ChaCha20-Poly1305.
- No address on chain: the recipient trial-decrypts every output with ivk. A wrong key is rejected by the tag.
- The ciphertext C_{out} is the sender's own copy, under ovk.

The doors



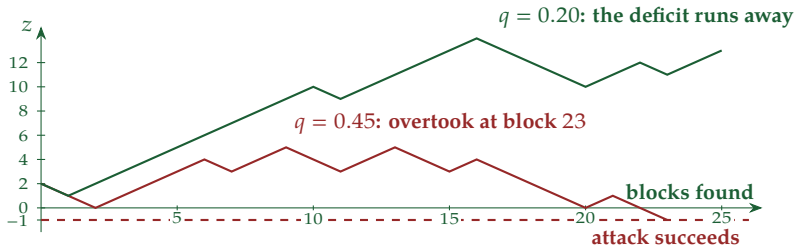
Every crossing is public in amount. Inside a pool, nothing is. Each pool's total is public and may never go negative.

What consensus adds

- A nullifier must not repeat within a transaction or across the valid chain.
- Transactions apply in block order: check $nf \notin \text{set}$, insert nf , append cmx .
- An anchor must be the final root of an earlier block — no spending an unmined note.
- A reorganisation rolls back tree, nullifier set and pool totals together.

The same shape as the UTXO set: an ordered, stateful check — over serial numbers that name no output.

Why one fork wins



Honest lead z ; each block moves it $+1$ with probability p , -1 with probability q .

$$a_z = \Pr[\text{lead ever reaches } -1] = (q/p)^{z+1} \quad (q < p)$$

here $p, q \in [0, 1]$ are hashrate shares, $p + q = 1$, not the primes.

Each block of deficit multiplies the chance by q/p : two forks cannot compete for long, and one quickly wins.

How Halo 2 proves

The claim becomes a table

“I know x with $x^3 + x + 5 = 35$ ” over $\mathbb{F}_{97}$; the witness $x = 3$:

$$3 \xrightarrow{\text{square}} 9 \xrightarrow{\times 3} 27 \xrightarrow{+3} 30 \xrightarrow{+5} 35.$$

One operation per row, one gate equation on every row:

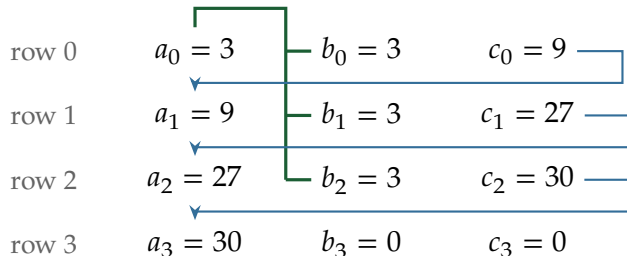
$$q_M a b + q_L a + q_R b + q_O c + q_C + \text{PI} = 0$$

every cell in $\mathbb{F}_{97}$.

row	advice			fixed					instance	
	a	b	c	q_M	q_L	q_R	q_O	q_C	PI	
0	3	3	9	1	0	0	-1	0	0	$x \cdot x = x^2$
1	9	3	27	1	0	0	-1	0	0	$x^2 \cdot x = x^3$
2	27	3	30	0	1	1	-1	0	0	$x^3 + x = s$
3	30	0	0	0	1	0	0	5	-35	$s + 5 = 35$

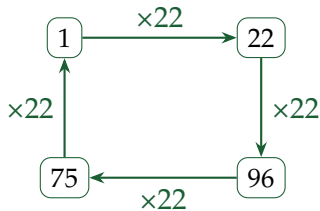
Advice: the prover's secret trace. Fixed: the circuit. Instance: the public 35. Copy constraints wire the rows; range checks are lookups into a table.

Equal values must be wired together



Equal numbers in a table are not a constraint. These *copy constraints* are enforced separately, by a permutation argument.

Give the rows multiplicative labels



$$\omega = 22, \quad \omega^2 = 96 = -1, \quad \omega^4 = 1$$

$$H = \{1, 22, 96, 75\} \subset \mathbb{F}_{97}$$

Row i gets the label ω^i : a cycle of field values.

Rows become polynomials

Row i lives at $\omega^i \in H$. Interpolate each column:

$$a(1) = 3, a(22) = 9, a(96) = 27, a(75) = 30 \implies a(X) = 90 + 61X + 22X^2 + 24X^3$$

$$G(X) := q_M ab + q_L a + q_R b + q_O c + q_C + \text{PI}$$

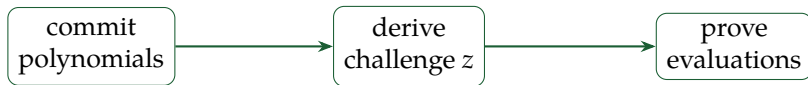
$a, b, c, q_\bullet, \text{PI} \in \mathbb{F}_{97}[X]$ of degree < 4 ; $G \in \mathbb{F}_{97}[X]$ of degree ≤ 9 .

$$Z_H(X) := \prod_{h \in H} (X - h) = X^4 - 1$$

$$\text{every gate holds} \iff G \text{ vanishes on } H \iff Z_H \mid G \iff G = t \cdot Z_H$$

A correct computation *is* a polynomial identity. A wrong one cannot be: the lie $x^2 = 10$ in row 0 leaves a remainder $24(1 + X + X^2 + X^3) \neq 0$.

One random point



1. The prover *commits* to a, b, c and t .
2. The challenge $z := H(\text{transcript})$ — Fiat–Shamir.
3. The prover opens $a(z), b(z), c(z), t(z)$.
4. The verifier checks one equation:

$$G(z) \stackrel{?}{=} t(z) Z_H(z)$$

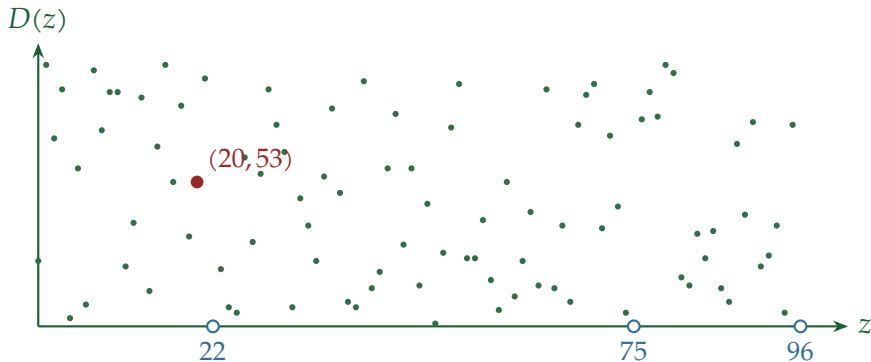
$z \in \mathbb{F}_{97}$ in the toy; $z \in \mathbb{F}_p$ in the protocol.

Honest, $z = 20$: $G(20) = 75 = 67 \cdot 46$. The row-0 lie: $G^*(20) = 20 \neq 64$.

A false identity is frozen before z is drawn; it survives only if z is one of its few roots.

A nonzero polynomial has few roots

The remainder $D(X) = 24(1 + X + X^2 + X^3)$ over $\mathbb{F}_{97}$: zeros at 22, 75, 96 only.



For uniform z this cheat passes with probability $3/97$; over the protocol's field, about 2^{-240} .

Committing to a polynomial

$$P := \langle \vec{a}, \vec{G} \rangle + [r] H, \quad \vec{a} = (a_0, \dots, a_{d-1}), \quad \vec{G} \text{ independent points}$$

$$a(z) = \langle \vec{a}, \vec{b} \rangle, \quad \vec{b} := (1, z, z^2, \dots, z^{d-1})$$

$a_i, r, z \in \mathbb{F}_p$; $\vec{G}, H, P \in \mathcal{U}$, the Vesta curve, whose scalar field is $\mathbb{F}_p$.

- A Pedersen commitment to the coefficient vector: hiding, binding.
- Opening at z is an inner product with a public vector.
- The problem: convince the verifier of $\langle \vec{a}, \vec{b} \rangle = v$ in fewer than d group elements, without sending $\vec{a}$.

The inner-product argument: one fold

Split every vector in half and publish two cross terms:

$$\begin{aligned} L &:= \langle \vec{a}_{\text{lo}}, \vec{G}_{\text{hi}} \rangle + [\ell] H + [\langle \vec{a}_{\text{lo}}, \vec{b}_{\text{hi}} \rangle] U, \\ R &:= \langle \vec{a}_{\text{hi}}, \vec{G}_{\text{lo}} \rangle + [r_R] H + [\langle \vec{a}_{\text{hi}}, \vec{b}_{\text{lo}} \rangle] U. \end{aligned}$$

Challenge u ; both sides fold:

$$\vec{a}' := u \vec{a}_{\text{lo}} + u^{-1} \vec{a}_{\text{hi}}, \quad \vec{G}' := u^{-1} \vec{G}_{\text{lo}} + u \vec{G}_{\text{hi}}, \quad \vec{b}' := u^{-1} \vec{b}_{\text{lo}} + u \vec{b}_{\text{hi}}$$

$$P + [v] U + [u^2] L + [u^{-2}] R = \langle \vec{a}', \vec{G}' \rangle + [r + u^2 \ell + u^{-2} r_R] H + [\langle \vec{a}', \vec{b}' \rangle] U$$

$$u \in \mathbb{F}_p^\times; L, R, U, H \in \mathcal{U}; \ell, r_R \in \mathbb{F}_p.$$

The same claim at half the length. The verifier folds $\vec{G}$ and $\vec{b}$ itself and never sees $\vec{a}$.

The recursion

After $k = \log_2 d$ rounds every vector has length 1. The proof: $2k$ points (L_j, R_j) , one scalar a , one blinder r^* .

$$P + [v] U + \sum_{j=1}^k ([u_j^2] L_j + [u_j^{-2}] R_j) \stackrel{?}{=} [a] G^{(0)} + [r^*] H + [a \cdot b^{(0)}] U$$

$a, b^{(0)}, r^*, u_j \in \mathbb{F}_p; G^{(0)}, L_j, R_j \in \mathcal{U}$.

- The scalar $b^{(0)} = \prod_j (u_j^{-1} + u_j z^{2^{j-1}})$, with u_1 the last challenge: logarithmic work.
- The point $G^{(0)} = \langle \vec{s}, \vec{G} \rangle$: one multi-scalar multiplication, shared across a batch of proofs.
- Binding: two openings would give a discrete-log relation among $\vec{G}, H$; an extractor rewinds three challenges per round.

Binding does not by itself give privacy

An unblinded evaluation is an equation in the secret coefficients:

$$f(z) = f_0 + f_1z + \cdots + f_{d-1}z^{d-1}.$$

random unused rows
mask the evaluations

random commitment blinders
hide the coefficients

Zero knowledge: the proof reveals nothing beyond the public statement.

The Orchard circuit

- Size: 2^{11} rows, 10 advice columns; arithmetic over $\mathbb{F}_p$ so Pallas points cm, rk, cv are native; commitments on Vesta.
- Gadgets: Sinsemilla (commitments, tree), Poseidon (nullifier), elliptic-curve arithmetic, lookups for range checks.
- The instance: $anchor, cv, nf, rk, cmx, flags$.
- Generators from a hash: no trusted setup, no ceremony.
- Proof size $2720 + 2272a$ bytes for a Actions: 4992 for one.

Every clause of the statement is a set of rows; one proof, one random point, one inner-product argument.

Back to the transaction

Proof	the hidden notes satisfy the statement
Spend signatures	the owners authorise this transaction
Binding signature	the committed amounts balance
Node state	anchor allowed; nullifiers unused

No single equation replaces the other checks.

The whole picture

check	Bitcoin	Ironwood
exists	UTXO lookup	Merkle path, inside the proof
not twice	delete the entry	nullifier, checked against the set
conserved	add the amounts	$\sum cv$, binding signature
authorised	script, signature	A6/A7, signature under rk

The validator learns that the four checks pass — and nothing about the note.